

Two Field Spread: numerical algorithms

v0.5.1 implementation and numerical assessment

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Status. The Markov solver, execution tape and long-run diagnostic correlations are implemented and tested. Numerical convergence and scientific acceptance remain pending; fractional transport and calendar subordination are future work. This draft describes the actual software; it does not amend the supplied theory paper or letter.

1 Model, agents and the inherited numerical construction

The state is two nonnegative densities, $\rho_B(x_j, u_n)$ and $\rho_A(x_j, u_n)$, in one market. A cell carries volume $\rho_i \Delta x$; x is log price. Fixed endpoint reservoirs are excluded from executable volume. Lit and latent sources aggregate supply; diffusion aggregates repricing; cancellation removes standing volume; symmetric reaction removes overlap. These are an aggregate closure with an order-level random-walk interpretation. The current solver does not simulate individual traders, wealth, beliefs or strategy choice.

The market-making sector is represented by pending completion stocks and source placement. All explicitly executed child volume in this pilot is attributed to that sector. Its opposite-side source is *completion-equivalent*: survival and execution probabilities are assumed inside the completion/placement specification. Source delivery is neither a new trade print nor independently observed settlement. Reaction loss is accounted for as a field sink, without inventing aggressor signs or prices for it. Thus the trade tape records the specified aggressor programme, not every possible microscopic match.

Let Q_B denote bid replenishment owed after sales and Q_A ask replenishment owed after purchases. The quote-setting sample obeys

$$Q_\Sigma = Q_A + Q_B, \quad Q_\Delta = Q_A - Q_B, \quad (1)$$

$$\Sigma = \Sigma_0 + \chi_s Q_\Sigma, \quad \mu = \mu_0 - \chi_m Q_\Delta, \quad q_b = \mu - \Sigma/2, \quad q_a = \mu + \Sigma/2. \quad (2)$$

Observed quotes p_b, p_a are side-density threshold crossings, not q_b, q_a . Zero signed inventory does not imply zero total pending exposure. Also, $\Sigma_0 = 12$ is imposed in this pilot: the existence of its baseline spread is not evidence of spontaneous spread formation from zero standoff.

Diana–Gebbie explicitly identify the need for two densities to represent a nonzero spread [1]. The correlation-emergence software supplies the operational DTRW, execution-record and observation conventions [2, 3]. Its two coupled books are not identified with the two sides of this market. The new density core implements the supplied spread paper’s Markov branch. For the common reaction-free interior operator, the prototype has

$$\rho^{n+1} = e^{-\nu \Delta u} \rho^n + \frac{r}{2} \delta_x^2 \rho^n + \Delta u S^n, \quad r = 2D\Delta u / \Delta x^2,$$

where $\delta_x^2 \rho_j = \rho_{j-1} - 2\rho_j + \rho_{j+1}$. Here cancellation is $(1 - \nu \Delta u) \rho^n$. Their difference is $(e^{-\nu \Delta u} - 1 + \nu \Delta u) \rho^n$, of one-step order Δu^2 . Matched transport agrees to roundoff at $\nu = \kappa = 0$; complete trajectories are not asserted identical. The prototype consumes before its field update; the present paper convention consumes after it. Phase labels preserve this distinction. See `compatibility-v0.4.1.json` for the executed comparison.

2 Algorithm 1: implemented Markov update

Inputs: previous post-event densities, post-event pending stocks P_B, P_A , fixed $\Delta x, \Delta u$, parameters and event-integrated requests. Both sides read one frozen field; new executions cannot complete immediately.

1. Deliver old pending cohorts: $C_i = (1 - e^{-\Delta u/\tau_h})P_i$ and $Q_i = P_i - C_i$. For the next-update control set $C_i = P_i$. This is the only “immediate” convention used. Form q_b, q_a from Q above.
2. Evaluate the paper’s lit and latent sources S_i^{ext} on the outward side of q_i . Normalize uniform nodal weights g_B on $[q_b - w, q_b]$ and g_A on $[q_a, q_a + w]$ so that $\Delta x \sum_j g_{i,j} = 1$. Reject empty or reservoir-truncated supports. No supply is placed between the placement edges.
3. For interior nodes, compute the pre-event field

$$\tilde{\rho}_{i,j} = (1 - 2a - \nu\Delta u - \kappa\Delta u\rho_{i,j})\rho_{i,j} + a(\rho_{i,j-1} + \rho_{i,j+1}) + \Delta u S_{i,j}^{\text{ext}} + C_i g_{i,j}, \quad a = D\Delta u/\Delta x^2.$$

Require $2a + \nu\Delta u + \kappa\Delta u\rho_{i,j} \leq 1$ everywhere. Keep endpoint reservoirs fixed; reject inadmissible states without clipping.

4. Read pre-event quotes by linear interpolation at density threshold ϵ : the rightmost downward bid crossing and leftmost upward ask crossing. Record crossing counts; reject a selected plateau, insufficient selected slope, reservoir failure or contact.
5. Execute each request in outward price priority, starting at its pre-event quote and ending at the configured execution depth. Restrict to interior node centres. Remove at most $\tilde{\rho}_{i,j}\Delta x$ per node; record actual fills, requests and unfilled demand. No reservoir is consumed. The current tape requires one aggressor side per event.
6. Let v_A be actual buy volume and v_B actual sell volume. Set $P_B^{\text{new}} = Q_B + v_A$, $P_A^{\text{new}} = Q_A + v_B$. Read post-event quotes and save both inventory samples, actual removals and separate source increments. An event at u is applied at the end of the interval ending at u ; pre/post fields share u with different phase labels.

Checks on every update. For each side, the interior volume change is boundary flux minus cancellation, minus symmetric reaction, plus external supply, plus completion, minus execution. The boundary increment is $D\Delta u(\rho_0 - \rho_1 + \rho_J - \rho_{J-1})/\Delta x$. Reaction loss is $\kappa\Delta u\Delta x \sum_j \rho_{A,j}\rho_{B,j}$ on each side. It cancels from imbalance and occurs twice in total density. Cumulative initiation must equal cumulative completion plus post-event pending stock, separately by side.

Initialization and recovery. Relax the same positive-reaction equations with empty pending stock and no executions to the configured residual. Recompute geometric completion weights when Δu changes, retaining physical τ_h . Preserve pending tails beyond a finite plotting horizon. Volume-balanced round trips need not restore liquidity at the quote at the same time: delivery occurs elsewhere and transport takes time. Neither the ledger nor a stationary field establishes dealer profitability or a strategic equilibrium. Hard source supports require grid-phase refinement; the current late spread offset has not been accepted as physical.

3 Algorithm 2: implemented execution tape

1. For event n retain nonzero executed node volumes $v_{n,j}$, parent identifier, aggressor sign ε_n (+1 buy, -1 sell), operational index and time. No fill means no trade observation. Preserve each fill row and the requested/unfilled amounts in the event record.
2. Following the prior executable convention, compute the primary execution log price

$$z_n = \frac{\sum_j v_{n,j} x_j}{\sum_j v_{n,j}}.$$

Store e^{z_n} as the geometric execution price and $\sum_j v_{n,j} e^{x_j} / \sum_j v_{n,j}$ as the separate arithmetic price VWAP. Do not call these identical. Store pre/post bid, ask, log midpoint and spread, and the placement and pending state that generated the event.

3. Preserve the true aggressor sign separately from the quote sign $\text{sign}(z_n - m_n^{\text{pre}})$ and tick sign $\text{sign}(z_n - z_{n-1})$. The first tick label is zero; a zero tick subsequently carries the last nonzero tick label, as in v2.0.0.
4. Using consecutive executed events, calculate $r_n^m = m_n^{\text{post}} - m_{n-1}^{\text{post}}$, $r_n^T = z_n - z_{n-1}$ and $\Delta s_n = s_n^{\text{post}} - s_{n-1}^{\text{post}}$. Retain absolute increments. First increments are missing, not zero. An instantaneous pre/post midpoint jump is stored separately from r_n^m , which also includes inter-event field evolution.

Prices are measured from actual density consumption; there is no independent synthetic price process. The current six same-sign children demonstrate the measurement path only. Their sign variance is zero, so no sign ACF or LMF claim is available from this run.

4 Algorithm 3: lagged correlation

The estimator and the registered long-run protocol below are implemented. The resulting correlations remain diagnostic pending numerical acceptance. For aligned finite series X, Y , define

$$\hat{C}_{XY}(k) = \frac{\sum_{n \in I_k} (X_n - \bar{X}_k)(Y_{n+k} - \bar{Y}_k)}{\sqrt{\sum_{n \in I_k} (X_n - \bar{X}_k)^2 \sum_{n \in I_k} (Y_{n+k} - \bar{Y}_k)^2}}.$$

Each mean is taken over its own overlapping slice. Positive k means X leads Y . Save pair counts, lag units and phase alignment. Zero variance is undefined, not zero correlation. Do not silently compress missing observations. This reproduces the v2.0.0 executable Pearson ACF. Its supplementary text instead prints a global-mean covariance normalization; that documentation mismatch is not carried into this supplement.

The long runs retain raw price paths and use $r^m, r^T, |r^m|, |r^T|$ for price ACFs. For a positive price, $|P| = P$; an absolute log-price level depends on the arbitrary origin. Spread-level ACFs require a checked stationary window. Price-level ACFs, if retained, must be labelled descriptive rather than evidence of long memory. A fixed valuation centre and fixed reservoirs can cause mean reversion; diffusive prices are not guaranteed.

5 Algorithm 4: registered flow and market-maker controls

No chartist or fundamentalist population is required for the current spread question. Static sources do not already encode strategies; fixed μ_0 is a valuation reference. The retained closure is unchanged.

1. For the finite programme, execute six children of volume 0.05 at $u = 1, 1.4, 1.8, 2.2, 2.6, 3$, either all buys or alternating buys/sells. Compare moving placement, next-update completion, both feedbacks off, $\chi_s = 0$ and $\chi_m = 0$. Use identical actual absolute fills. Show total and signed pending stocks separately; delivery is not a second trade.
2. For long runs, draw integer parent length $L = \min(\lfloor 2(1 - U)^{-1/1.5} \rfloor, 512)$ with U uniform on $[0, 1)$. Draw independent centred parent signs. Sample the initial length from $\Pr(L_0 = l) = l \Pr(L = l) / E[L]$, with uniform integer age $0, \dots, l - 1$. Keep distinct parent IDs even when adjacent signs agree.
3. Execute children of volume 0.002 every 0.01 operational units. Use seeds 41001–41008, 2048 burn events and 8192 retained events per path. Compare moving placement with the same splitting tapes at fixed placement, and independent signs with seed offset 100000 and matched volume/rate. The first retained increment uses the preceding burn-event price.
4. Compute ACFs within each path for $\varepsilon, r^m, r^T, |r^m|, |r^T|, s$. Compute CCFs for (ε, r^m) , $(\varepsilon, \Delta s)$, (r^m, r^T) , $(r^m, \Delta s)$, $(|r^m|, s)$ and $(|r^T|, s)$ through lag 128. Average eight independent paths; bands are descriptive mean $\pm$ twice the between-path standard error. Paired controls are not extra replicates. Save pair counts, per-path correlations and spread half-window means.

The exact stationary single-active-parent sign reference is

$$C_\varepsilon(k) = \frac{E[(L - k)_+]}{E[L]}.$$

For an untruncated tail with $1 < \beta < 2$ this renewal benchmark has exponent $\beta - 1$, consistent with the order-splitting motivation [4]. The implemented capped law is not the full concurrent-parent LMF population, and cannot establish asymptotic long memory. The configuration reserves lags 8–64 for a possible fit; no exponent has been fitted or accepted. Persistent signs are declared input, not memory generated by density diffusion.

Grids and retained data. The main finite figures use $(\Delta x, \Delta u) = (0.025, 0.00025)$; F4 repeats ten cases also at $(0.05, 0.001)$ and three balanced cases at $(0.0125, 0.0000625)$. Long-run groups use $(0.05, 0.001)$ and the first two splitting seeds repeat at $(0.025, 0.00025)$. Symmetric domains are $[-12 - \Delta x/2, 12 + \Delta x/2]$, placing initial source edges between nodes. All 26 long paths retain actual fills, parent identities and pre/post quotes. All completed without material unfilled demand; no failed path was silently selected out.

Figure inventory. F2: density snapshots; F3: prices and market-maker state; F4: matched mechanisms and flow; V1: held-state profile video. F5 has aligned execution-price, spread and sign paths plus the six ACFs. F6 has the six CCFs. Every panel uses retained simulation data or the declared sign-input reference. No independent theory gallery is required.

6 Assessment and limits of v0.5.1

The 18-case study varies grid, time step, domain, support phase, threshold and execution depth while retaining the supplied equations. All 25 focused tests pass. Main finite trajectories have field and completion-ledger errors below 3.2×10^{-15} and 2.6×10^{-15} ; this verifies accounting, not convergence. The source and execution supports remain nodal and hard.

The node-aligned late spread change at $u = 8$ falls from 0.19674 to 0.02642 as Δx decreases from 0.1 to 0.0125. Between-node source placement reduces the fine-grid offset to 0.00496. These are discretization and finite-horizon effects, not permanent impact. The finest paired spread differences remain 0.02059 on-node and 0.01224 between-node, exceeding the registered 0.01 tolerance. Separate domain checks assess the small reservoir displacement from half-cell padding. Execution depths 1 and 4 produce identical paths; source smoothing was not introduced.

The balanced programme leaves the same total post-event pending stock as the directional programme after the sixth child, approximately 0.1379, with signed stock approximately 0.0272 in magnitude. Thus balanced flow does not empty the market-maker exposure. The balanced centre-only maximum midpoint effect is 0, 0.01494 and 0.00737 on successive grids. At the coarse grid the moving centre does not change source-node membership. It would be incorrect to infer from that zero that centre feedback is absent.

The v0.5.1 comparison holds the same two full tapes and physical child volume fixed. It uses $\Delta x = 0.025, 0.0125$ at common $\Delta u = 0.0000625$; the existing $\Delta x = 0.025, \Delta u = 0.00025$ paths provide the time control. For pre/post-execution threshold midpoints,

$$\Delta m_n = (m_n^+ - m_n^-) + (m_n^- - m_{n-1}^+).$$

Actual fill counts and support-membership changes at every field update are recorded without feeding back into the recurrence. The maximum correlation changes over the two pairs and retained lags are:

Refinement	Midpoint ACF	Spread ACF	Any CCF
Time only	0.00389	0.00547	0.01130
Space only	0.10340	0.12893	0.11826

Every retained child fills one node on the coarser grid and two on the finer grid. Source support changes within approximately 11–12% versus 24–25% of event intervals. Execution-jump standard deviation falls from about 0.0182 to 0.0152; its cancellation with between-event movement also changes. This identifies spatial support/execution resolution as the main remaining correlation sensitivity, without assigning a unique cause. Time refinement still gives occasional spread differences up to 0.02461; small ACF changes do not establish pathwise quote accuracy. Fine-grid lag-one midpoint ACF is 0.636–0.651, so whitening is not demonstrated. Trade-return ACF changes are below 0.000175 in the spatial pairs. Eight benchmark paths and two refinement pairs establish neither universal price statistics nor asymptotic long memory. Numerical acceptance remains open.

Square-root impact and equilibrium. A programme of one volume supplies no impact exponent. A future spread/meta-order study needs matched baselines, volume/duration/participation controls and an applicable limit [5]. The separated two-front model does not automatically inherit the single-front limit. Positive Σ_0 is imposed; source completion does not establish a second executed leg or financial equilibrium.

7 Future memory and clock compatibility

Keep four mechanisms distinct: event-time order splitting (β), operational transport memory (α_u), round-trip completion (h, τ_h) and calendar observation/subordination (α_t). Give stochastic components separate recorded seed streams. The event/operational/calendar distinction also follows the companion account [6]; its abstract provides context here, not verification of a fractional two-side solver.

The prior raw Sibuya kernel is

$$K_1 = \alpha_u, \quad K_2 = \frac{1}{2}\alpha_u(\alpha_u - 1), \quad K_k = K_{k-1} \left(1 - \frac{2 - \alpha_u}{k}\right), \quad k \geq 3.$$

The prototype applies survival $e^{-\nu(k-1)\Delta u}$ to lagged transport and $e^{-\nu\Delta u}$ to the current survivor, once each. Cancellation is not already folded into the raw kernel. $\alpha_u = 1$ leaves only $K_1 = 1$. Its scaling is $D_{\alpha_u} = r\Delta x^2/(2\Delta u^{\alpha_u})$; changing α_u while silently retaining an incompatible $r, \Delta u, D$ is invalid.

The current state has no transport-history buffer. A future fractional two-side model must derive the survival and history treatment of nonlinear reaction, cancellation, births and consumed volume. It must not resurrect removed liquidity through old history. This requires a stated formulation and tests against the prior primitive and its Markov limit; inserting the kernel into the existing Laplacian alone is insufficient.

The paper's stationary placement construction requires a finite mean completion lag. A heavy-tailed transport waiting time is not permission to replace the completion density with an infinite-mean law. That would change the equilibrium assumptions and can make pending exposure diverge under sustained input.

The correlation-emergence observation layer acts on completed operational paths. A future projection must select the previous completed state under the declared clock, retain index maps and reject out-of-support queries. Do not interpolate across an execution. Use a coherent book observation for bid, ask and midpoint before testing separate reporting delays. A held last trade price creates neither another trade nor another sign. Empty calendar bins have zero signed flow, not a repeated aggressor; distinguish them from missing pre-first-trade observations. Compare shocked/control paths using the same realized clock. These rules are specified, not yet implemented.

References

- [1] D. Diana and T. Gebbie, *Non-uniformly sampled simulated price impact of an order-book*, Journal of Computational and Applied Mathematics **456** (2025), 116202. doi:10.1016/j.cam.2024.116202.
- [2] C. N. Angstmann and T. Gebbie, *Correlation emergence and the Epps effect in two coupled limit order books*, arXiv:2606.14182, version 3.
- [3] T. Gebbie, *correlation-emergence-reproducibility*, v2.0.0 and v2.2.0. Exact commits and inspected component hashes are in SOURCES.md.
- [4] F. Lillo, S. Mike and J. D. Farmer, *A theory for long-memory in supply and demand*, Physical Review E **71** (2005), 066122. arXiv:cond-mat/0412708.
- [5] J. Donier, J. Bonart, I. Mastromatteo and J.-P. Bouchaud, *A fully consistent, minimal model for non-linear market impact*, arXiv:1412.0141.
- [6] C. Angstmann and T. Gebbie, *Event-Time Order-Flow Memory, Operational-Time Impact, and Subordinated Market Observables*, arXiv:2609.13715 (2026).